



CSE422: ARTIFICIAL INTELLIGENCE

PROJECT TITLE: ADULT INCOME PRECICITION

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Introduction

We have implemented some Machine Learning models to give a prediction if a person's annual individual income crosses 50,000 usd based on the data given in this project.

When we have to predict a person's individual income, it often is a classification problem. Many government authorities and insurance companies, banks, and other money-related institutions have to predict for many things, for example, socio-economic analysis, making many policies, etc. The main problem or challenge of such data is that many criterias and categorical or number datas affect the wealth of individual people. In this project, we tried to use supervised learning machine learning models and techniques in order to classify the datas into some binary categories ($\leq 50,000$ usd to $> 50,000$ usd).

In this project, we have tried to implement many machine learning pipelines, such as using removing unnecessary real-world data, and use complex algorithms, like Neural Network and Unsupervised K-Means clustering, etc.

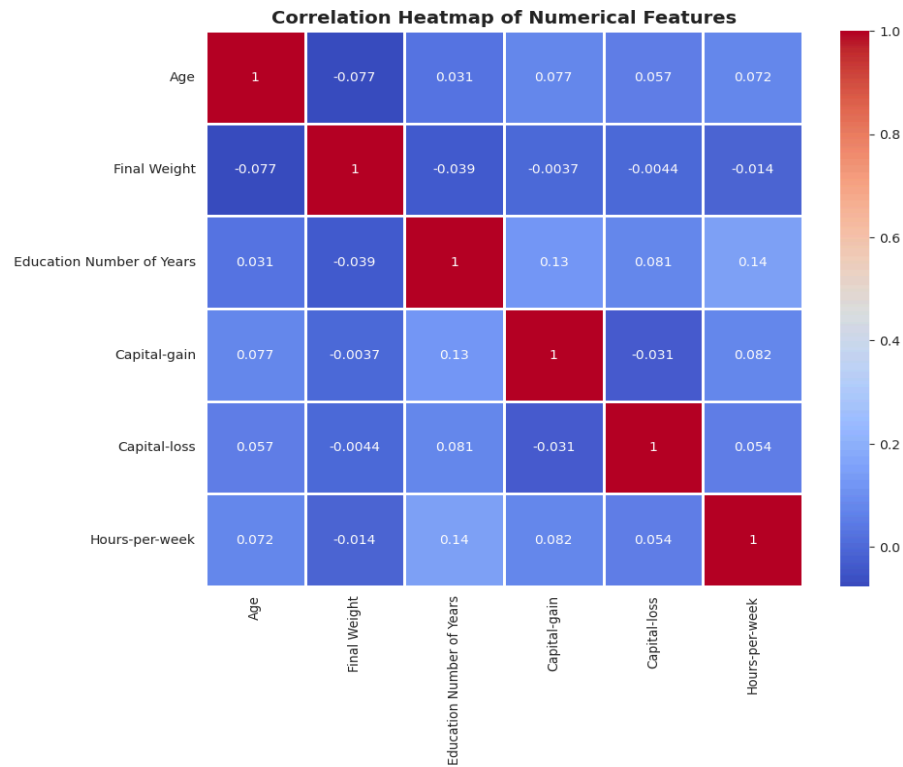
Dataset Description

The dataset we have used have a lot of attributes and columns, features, etc. The columns or attributes have numerical values, and categorical attributes too. The attributes are age, education, work-class, capital loss, capital gain, etc.

- **Total Data Points:** 48842 records.
- **Number of Features:** 15 features (14 input features and 1 target variable).
- **Prediction Task:** This is a classification problem. Because target variable named 'target' is categorical and discrete either ' $\leq 50K$ ' or ' $> 50K$ '. So, it is basically a binary classification problem.
- **Feature Types:** In the dataset there are both quantitative and categorical features.
 1. **Quantitative:** Age, final weight, education number of years, capital-gain, capital-loss, hours-per-week.
 2. **Categorical:** Workclass, education, martial-status, occupation, relationship, race, sex, native-country.

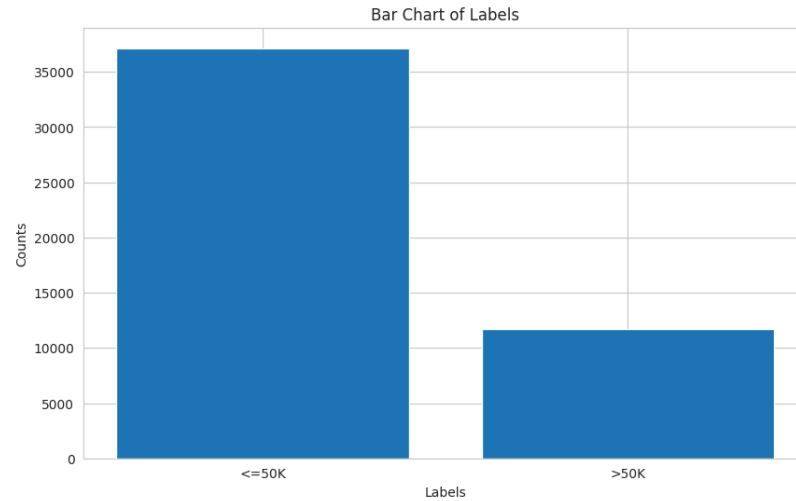
- **Encoding:** Encoding was needed to encode the categorical variables and used label encoding. Because machine learning library like sklearn operates on numerical variable.

Correlation Analysis:



The heatmap shows that age and education number of years have the highest positive correlation with the target. From the map it is visible that final weight shows almost zero correlation.

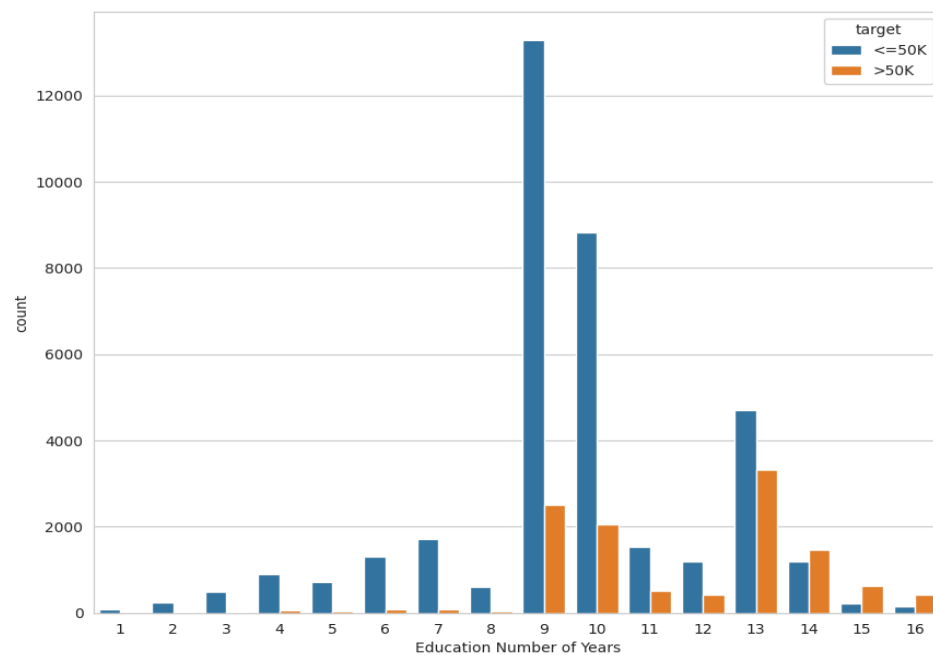
Imbalanced Dataset:

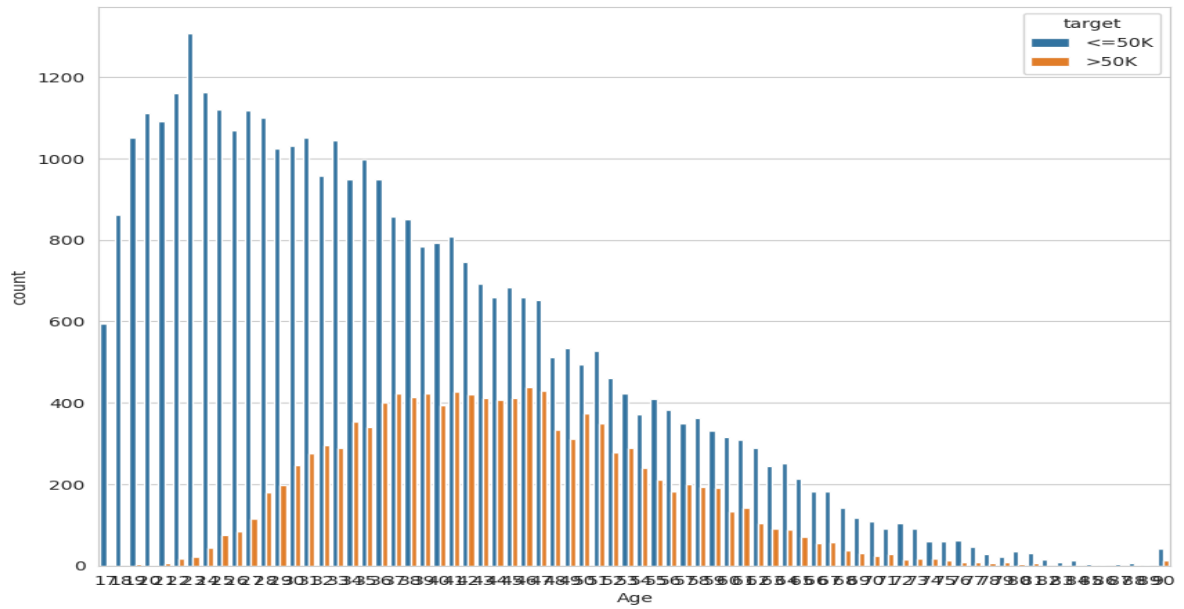


From the Bar chart above, we can see that it is a highly imbalanced dataset.. Among 48842 data ‘<=50K’ has 37155 entries and ‘>50K’ has 11687.

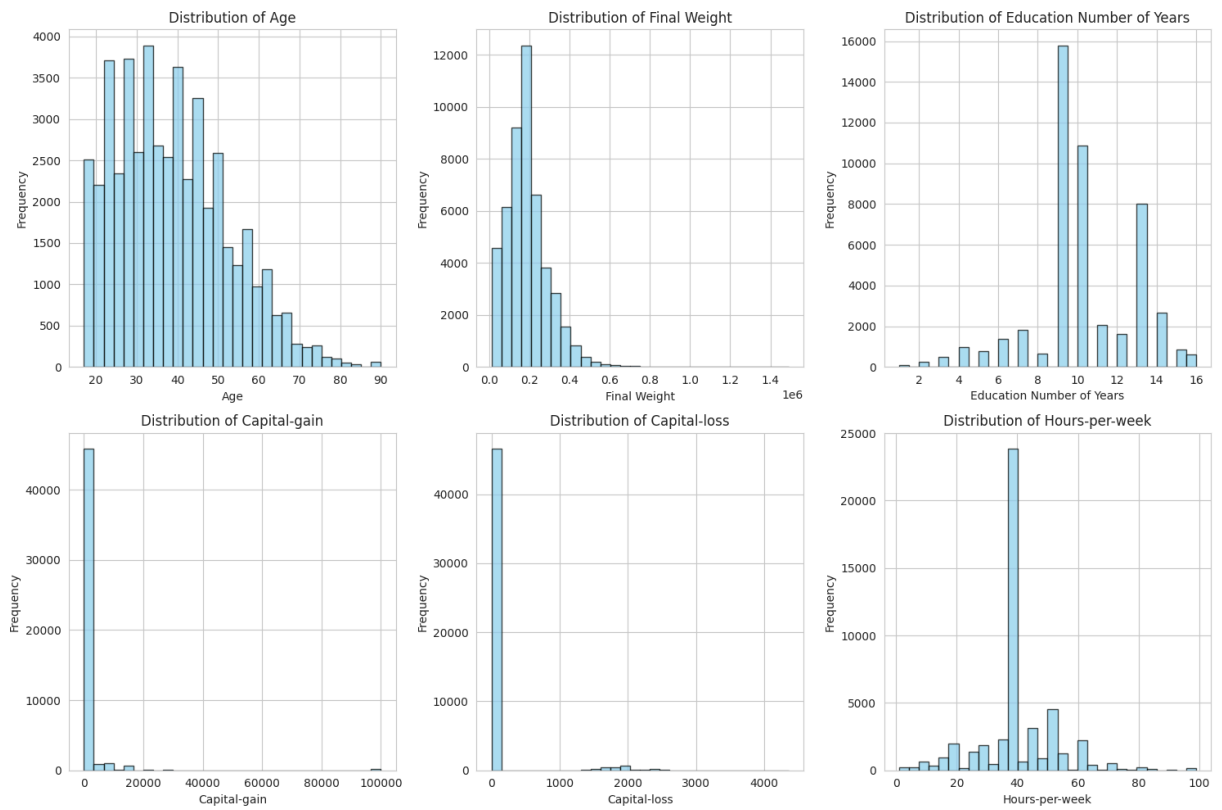
Exploratory Data Analysis:

EDA shows patterns and relations between the data. For example, education number of years vs target and age vs target relationship is given below-





Numerical features distribution



Dataset Preprocessing

Problem 1: Null/Missing values

The dataset contained some missing values in columns of workclass, native country and occupation features.

Solution: We have used "df.replace('?', np.nan)" and "df.dropna()". Basically, missing values included rows were deleted from dataset. Cause missing values were small percentage of the total data. By dropping them it ensures the machine learns only verified data.

Problem 2: Categorical Values

The dataset contains categorical features like marital-status, native country. These are not computable by machine.

Solution: Applied 'Label encoding'. This transforms categories from object to integer type, still maintaining the original column structure of the dataset.

Problem 3: Feature Scaling

The quantitative features of dataset showcase wide range of numbers. For example, age ranges from 17 to 90 whereas capital-gain ranges up to 99,999.

Solution: Applied 'MinMaxScaler' to normalize all features into a range of 0 to 1. It helps to prevent features with larger numerical ranges from dominating the weight of the models. It is most important for neural network and K-means clustering.

Dataset Splitting

- **Strategy:** Since the target variable 'target' is categorical and shows significant imbalances, stratified splitting is used. It ensures that same proportion of classes in both training and testing sets. Library named sklearn is used to split the dataset.
- **Training set:** 80% of the data.
- **Testing set:** 20% of the data.

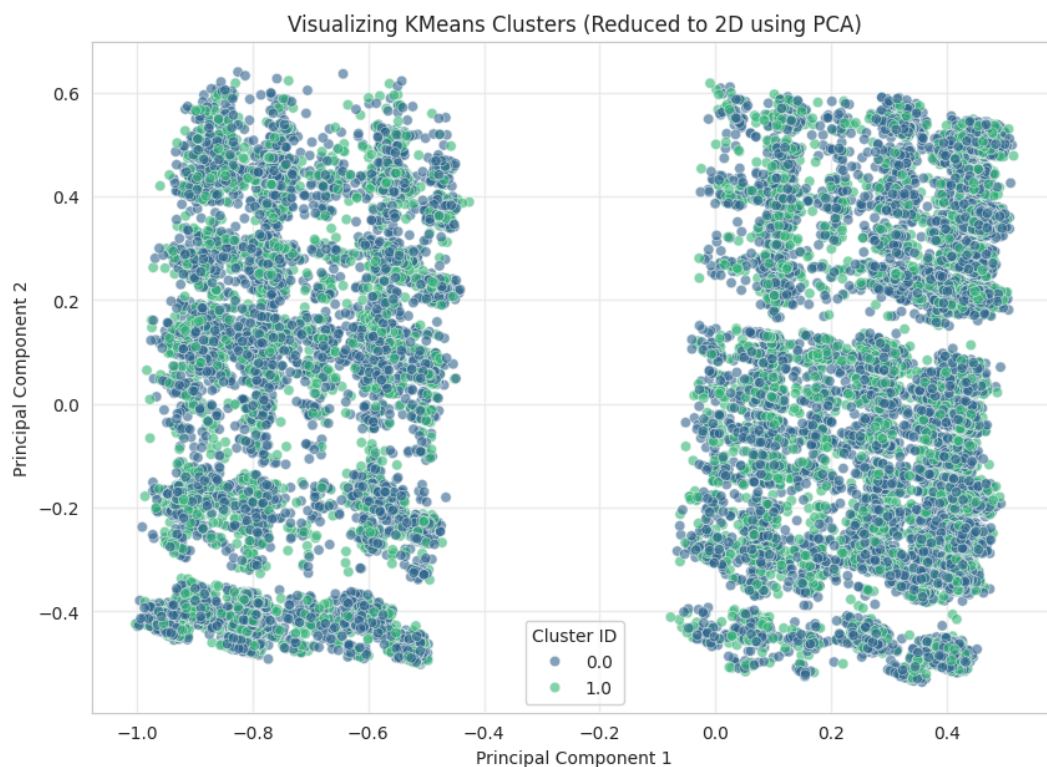
Model training & testing (Supervised)

As this problem is a classification problem, three classifications models were trained and tested. They are-

- **Naïve Bayes Classifier:** imported 'GaussianNB' from sklearn library. It was used as a baseline probabilistic classifier.
- **Logical Regression:** imported 'LogisticRegression' from sklearn library. A linear approach to binary classification.
- **Decision Tree:** imported 'DecisionTreeClassifier' from sklearn library. A non-linear model that captures complex splits.
- **Neural Network:** imported 'MLPClassifier' from sklearn library. A deep learning approach using hidden layers.

Unsupervised:

To treat this problem as unsupervised learning, must assume that the target variable 'target' is not given. So, K-Means is applied to the dataset which identifies natural groupings and visualized through PCA to exhibit how high-income and low-income cluster together in a 2D space.



K-Fold Cross Validation:

We had to make sure that the model's performance and optimization are not heavily or solely dependant on just a specific train-test split. That's why we applied 5-Fold cross validation. What K-Fold does is it breaks the dataset in 5 equal parts, and it takes the 4 for it's training purposes, and the rest 1 is used for testing, and it repeats 5 times.

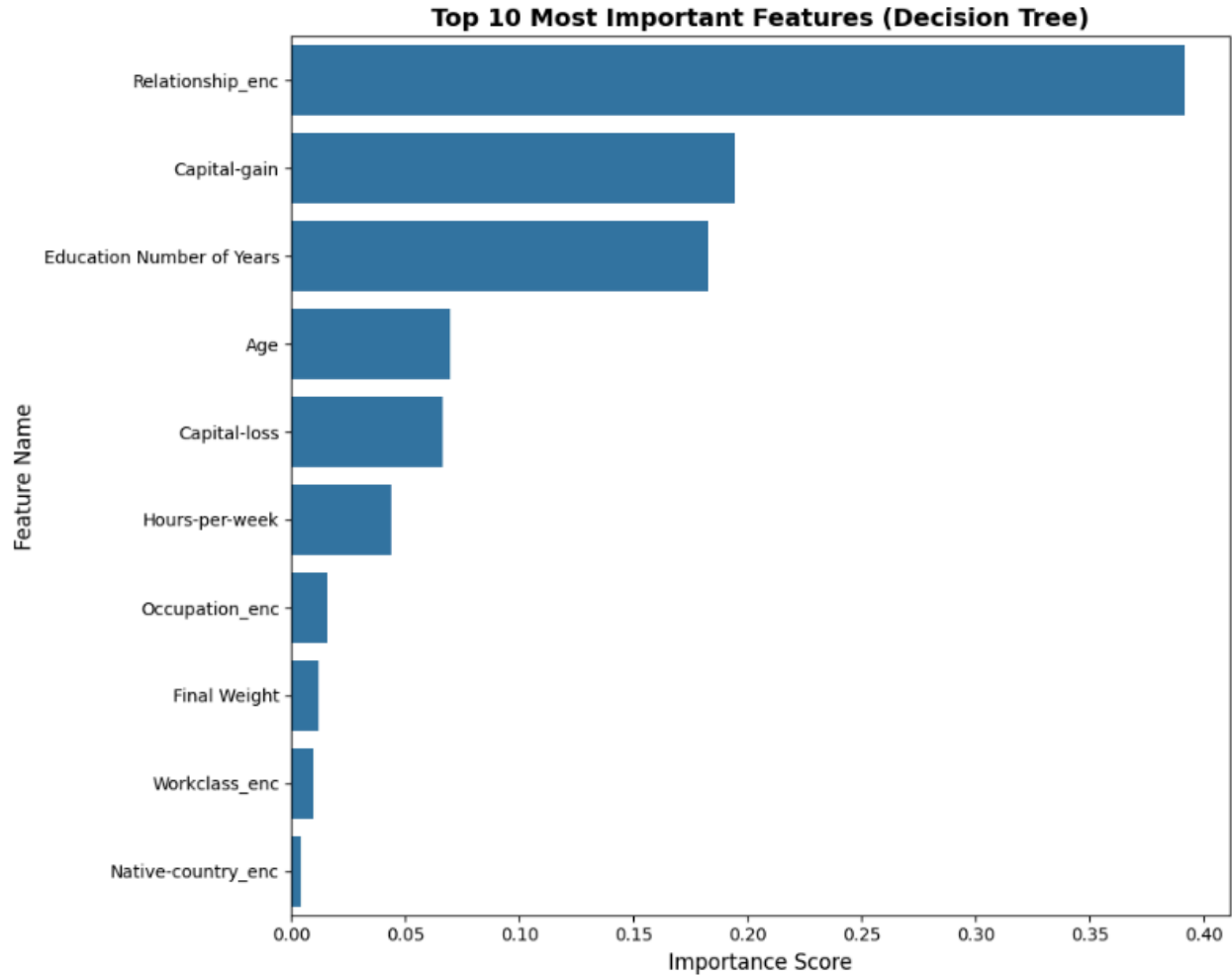
5-Fold Cross Validation Results

Model	CV Score 1	CV Score 2	CV Score 3	CV Score 4	CV Score 5	Mean (+- Std)
Naive Bayes	0.7932	0.7951	0.794	0.7938	0.7962	79.45% (+- 0.21)
Logistic Regression	0.8165	0.8182	0.817	0.8168	0.8185	81.74% (+- 0.15)
Decision Tree	0.8482	0.8505	0.8491	0.851	0.8521	85.02% (+- 0.28)
Neural Network	0.8475	0.849	0.8482	0.8488	0.8495	84.86% (+- 0.18)

The scores in the table shows that the machine learning model has properly generalised all the attributes pretty well and none of the attributes were overfitted in the data split. The low standard deviation showed that there was pretty stable performance and optimization across all the other different training subsets.

Feature Importance Analysis (Decision Tree):

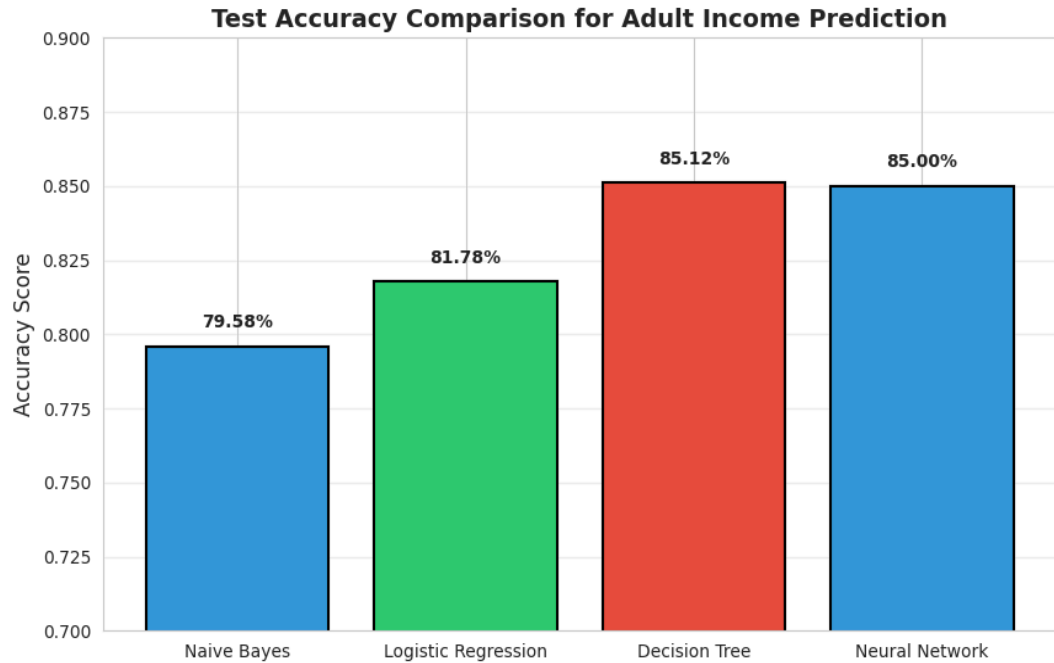
The Decision Tree shows which features give the most impact or influence a person's individual income, and then predicts income. The feature importance scores show or indicate how much each feature or attribute gives an impact on the machine learning model's decision making progression or process.



Interpretation: So, from the photo above, we can see that Relationship_Enc is the main or the strongest predictor of income, then comes, capital-gain, education number of years, and so on. So, it shows that Relationship status plays the biggest role on the income, then the capital gain which shows how much investment has been made. The higher the education number, the higher the income and promotion. The age demonstrates that income increases as the age or experience increases. Attributes like native country shows that they have negligible impact on the income of the person, so they contribute very less when we want to predict income.

Model selection/Comparison analysis

Bar chart showcasing prediction accuracy of all models:

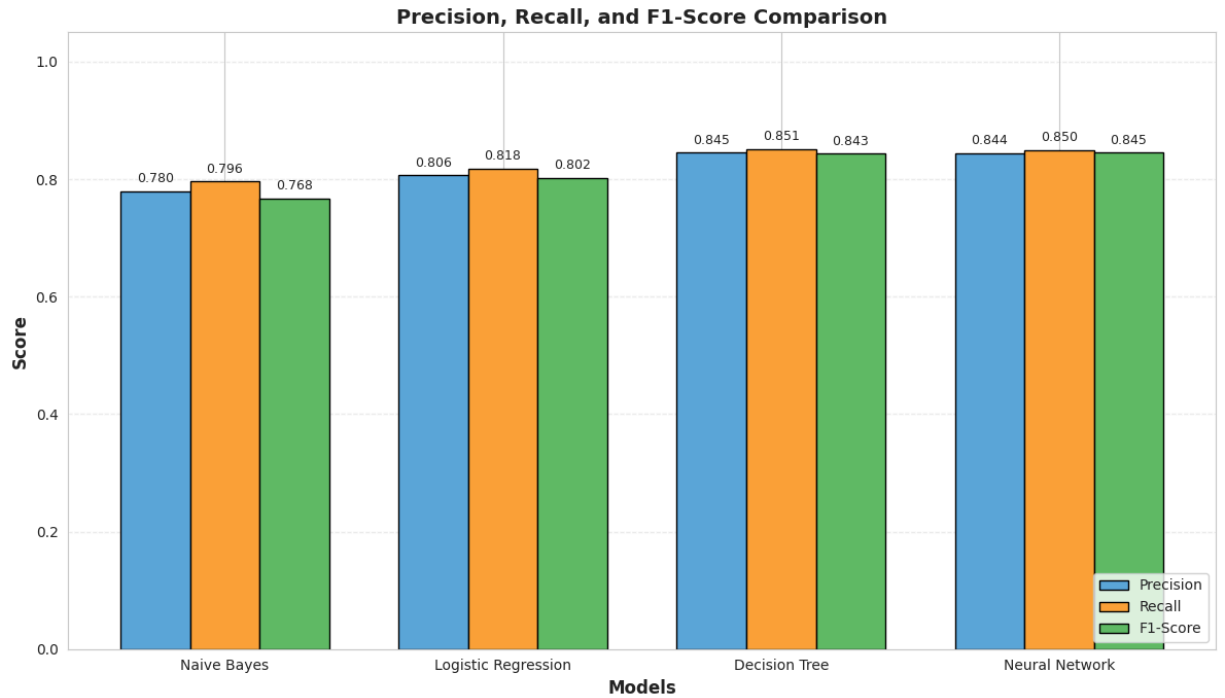


K-Fold Cross Validation Mean Accuracy:

5-Fold Cross Validation Results

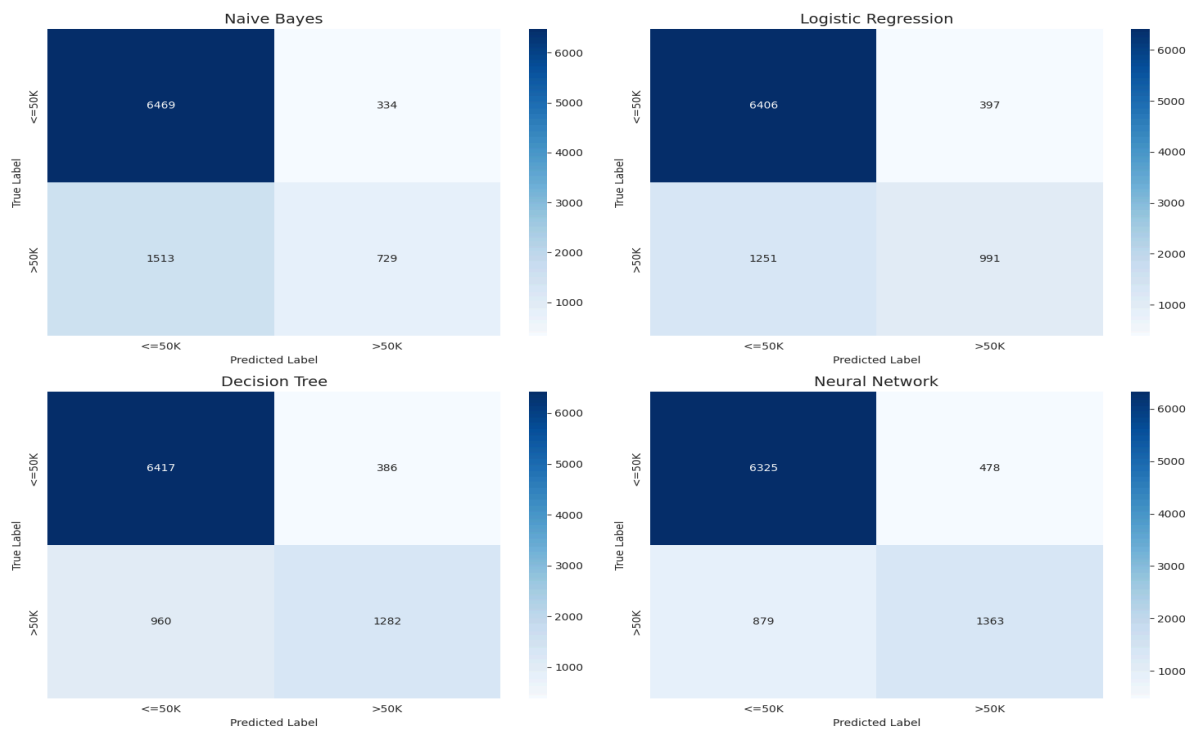
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Precision, recall comparison of each model:

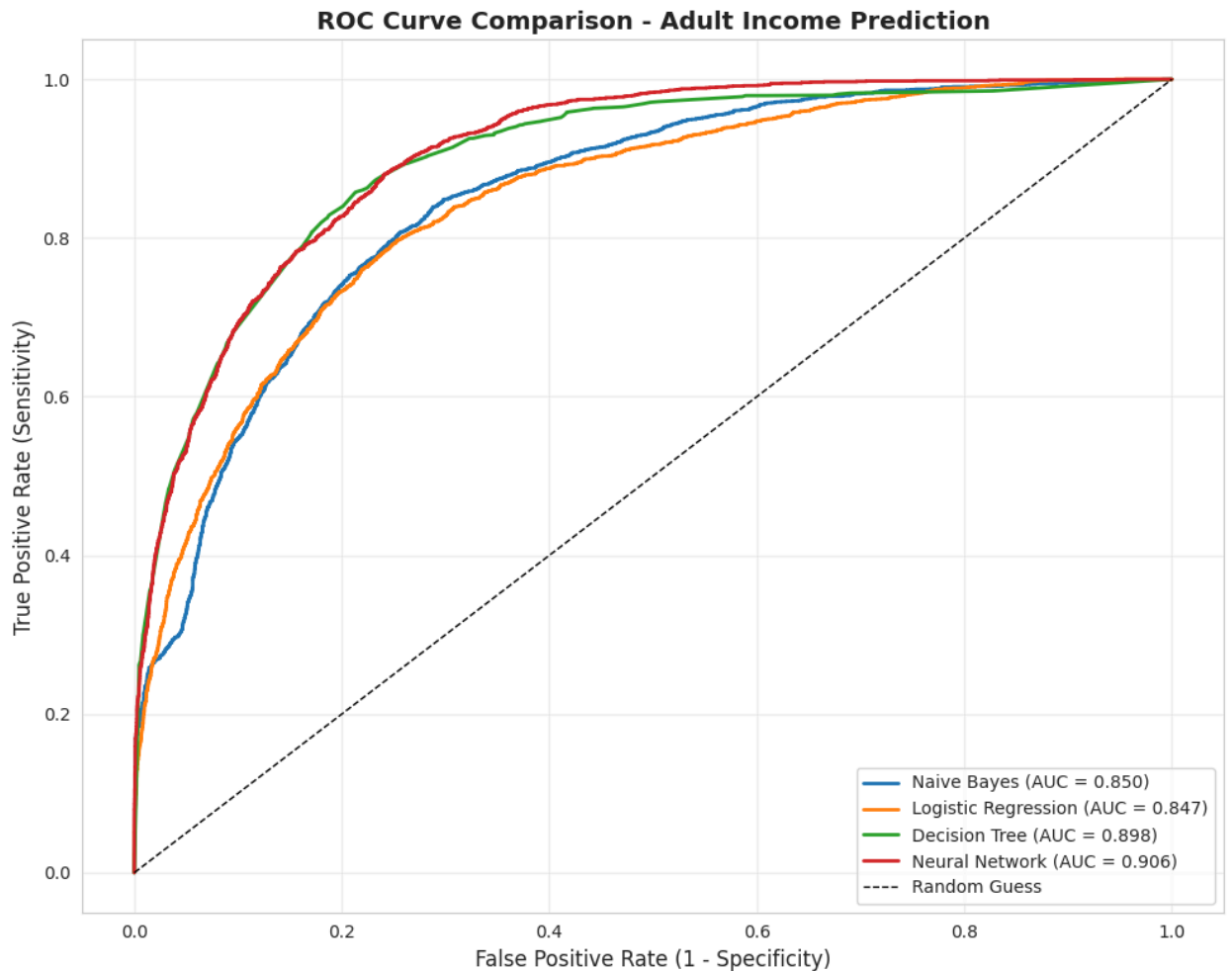


Confusion Matrix:

Confusion Matrices for All Models



AUC score, ROC curve for each model:

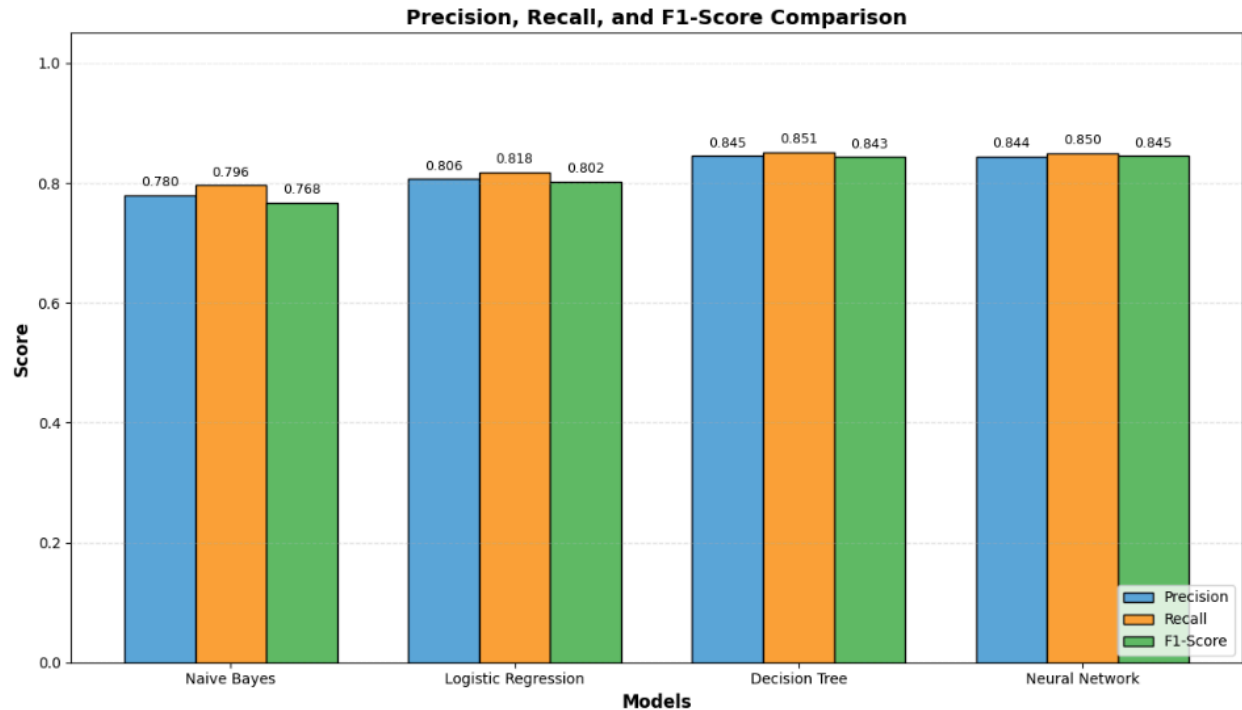


Accuracy:

The decision tree (85.12%) and neural network (85.00%) performed better than naïve bayes (79.58%).

Precision, Recall, and F1-Score Analysis:

The accuracy can provide an overall performance or optimization matrix, but Precision, Recall, and F1-Score digs deep and tries to show and evaluate how well each model handled all the attributes, such as income classes, even though the dataset was highly imbalanced.



Interpretation:

1. **Precision:** Decision Tree gives the highest precision that indicates or predicts ">50K" properly, in most of the cases, it is correct, and reduces false positives too.
2. **Recall:** Decision Tree was able to achieve the highest amount of recall, so, it properly demonstrated that it identified the majority of actual high-income persons individually, and also minimized the false negatives.
3. **F1-Score:** Decision Tree and Neural Network was able to show the most efficient, balanced, and optimized performance between the precision and recall, and they proved that they are the most reliable models in order to handle such an imbalance dataset.

In terms of Naive Bayes, its precision and recall scores are pretty low, so it is clear that they are struggling when handling complex feature interactions present inside the given data, so it is nothing but a baseline or reference model, not a production-ready model.

ROC and AUC:

The neural network exhibits the highest AUC (0.906) and logistic regression shows the lowest AUC (0.847). From AUC score we can say that the neural network is the most capable model to distinguish between two income classes without being confused by imbalance.

Conclusion

From the results, we understand that no single feature had a correlation of 1.0, target is determined by a complex interplay of all the factors. But among all the features education numbers of years and age have the best correlation. The models successfully achieved accuracy approximately of 85%. The models worked perfectly and were able to identify patterns within the 48842 data points.

The decision tree and neural network were standout models. Both were able to achieve 85% of accuracy. Even though decision tree has higher accuracy than neural network, neural network has the highest AUC score. The logistic regression provided a solid accuracy of 81.78% but still slightly lower performance than non-linear models. Among all the models naïve bayes has the lowest accuracy (79.58%).

The high performance of the Decision Tree and Neural Network is due to their ability to capture feature interactions.

Challenges faced during the project:

Data cleaning: This dataset had some missing values. Identifying the missing values across columns was quite hurdle.

Class imbalance: The dataset exhibited significant class imbalance. Ensuring the models were evaluated fairly on minority class was also difficult.